Inequalities: When Does the Symbol Flip?
Answer Key

Grades 6–7 Expressions and Equations

Quick Answers

1. 8
2. $-\infty$, -3, ¡built-in function open¿
3. -3, ∞, *closed*
4. -6, ∞, ¡built-in function open¿
5. 18
6. $-\infty$, -5, ¡built-in function open¿
7. $-\infty$, -3, *closed*
8. —

Curriculum

Level: Grades 6–7 Expressions and Equations
7.EE.B.4 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

1. *If they got -8: kept the sign of the first factor: read -4 times -2 as -8*
5. *If they got -22: multiplied -5 by -4 and wrote -20, losing the negative-times-negative rule*

1. Multiply both sides of $6 > -2$ by -4.
Solution. A negative times a negative is positive, so the right side is $-4 \times (-2) = \boxed{8}$ The left side is $-4 \times 6 = -24$. The new pair is -24 and 8, and $-24 < 8$, so the true statement is $-24 < 8$: the original "$>$" became "$<$". Multiplying both sides by a negative reversed the symbol. *Common wrong answer:* -8, from keeping the sign of the first factor.

2. Solve $-4x > 12$.
Solution. Divide both sides by -4. Because -4 is negative, the symbol reverses on that step:

$$\frac{-4x}{-4} \overset{\text{flip}}{<} \frac{12}{-4}$$

$\boxed{x < -3}$ *Check:* $x = -10$ gives $-4(-10) = 40 > 12$ ✓, and $x = 0$ gives $0 > 12$, which is false — exactly what a solution set of $x < -3$ predicts.

3. Find and fix Priya's work on $-7x \leq 21$.
(a) Test her answer. Priya claims $x \leq -3$, so $x = -10$ should work. Substituting into the original: $-7(-10) = 70$, and $70 \leq 21$ is *false*. So her solution set contains numbers that are not solutions.
(b) The error. She divided both sides by -7 but kept the "$\leq$" pointing the same way; dividing by a negative reverses the symbol.
(c) Correct solution. $\frac{-7x}{-7} \overset{\text{flip}}{\geq} \frac{21}{-7}$, so $\boxed{x \geq -3}$ *Check:* $x = 0$ gives $-7(0) = 0 \leq 21$ ✓.

4. Solve $-\dfrac{x}{3} < 2$.
Solution. The coefficient of x is $-\frac{1}{3}$. Multiply both sides by -3; that is a negative multiplier, so the symbol reverses:

$$(-3)\left(-\frac{x}{3}\right) \overset{\text{flip}}{>} (-3)(2)$$

$\boxed{x > -6}$ *Grading note:* a student who multiplies by 3 first (getting $-x < 6$) and then by -1 (getting $x > -6$) is also correct — the flip happens on the -1 step. Only one of the two moves is a negative multiplier.

5. Check the boundary of $-5x - 2 \le 18$ at $x = -4$.
Solution. Substitute and evaluate, minding the double negative: $-5(-4)-2 = 20-2 = \boxed{18}$ The expression equals exactly 18, so $x = -4$ is the boundary and the solution set is $x \ge -4$ (dividing $-5x \le 20$ by -5 flips the symbol). *Common wrong answer:* -22, from writing $-5(-4)$ as -20.

6. Solve $4 - 3x > 19$.
Solution. Subtract 4 from both sides — subtraction never changes the symbol:

$$-3x > 15$$

Now divide by -3, which does:

$$\frac{-3x}{-3} \overset{\text{flip}}{<} \frac{15}{-3}$$

$\boxed{x < -5}$ *Check:* $x = -10$ gives $4 - 3(-10) = 34 > 19$ ✓.

7. Find and fix Dev's work on $8 - 2x \ge 14$.
(a) Which step is wrong. Step 1 ($8 - 2x \ge 14 \to -2x \ge 6$) is fine: subtracting 8 from both sides is a shift, and shifting both numbers by the same amount cannot change which one is larger. Step 2 is the error — he divided by -2 without reversing the symbol.
(b) The counterexample. $x = 0$ satisfies Dev's answer $x \ge -3$. In the original, $8-2(0) = 8$, and $8 \ge 14$ is false. So $x = 0$ is in his solution set but is not a solution.
(c) Correct solution. $\dfrac{-2x}{-2} \overset{\text{flip}}{\le} \dfrac{6}{-2}$, so $\boxed{x \le -3}$ *Check:* $x = -10$ gives $8-2(-10) = 28 \ge 14$ ✓.

8. Manual review — expected reasoning. Adding the same number to both sides slides both numbers the same distance in the same direction, so their order is unchanged: if $2 < 5$ then $2 + (-5) = -3$ and $5 + (-5) = 0$, and $-3 < 0$ — still "less than".
Multiplying by a negative does something different: it reflects the number line through 0. Points that were on the left land on the right. If $2 < 5$, then $2 \times (-5) = -10$ and $5 \times (-5) = -25$, and now $-10 > -25$. The number that was smaller ends up larger, which is exactly what "the symbol reverses" records.
Accept any answer that makes the ordering point: a shift preserves order, a reflection across zero reverses it (or, equivalently, "the farther a positive number is from zero, the farther its negative is from zero on the other side"). Do not accept "because the rule says so" or an answer that only restates the procedure without reference to order or position.